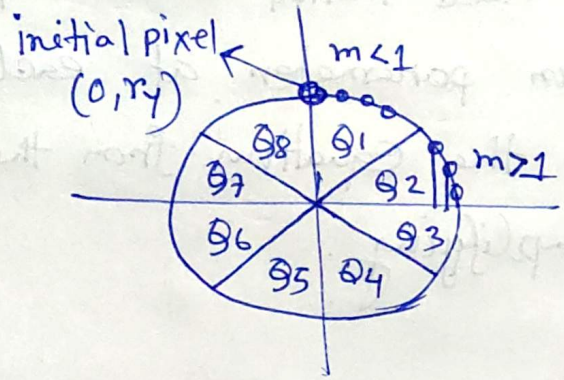
Ellipse



in Q1

x incremented but y changes a little

x independent and y dependent variable.

in Q2

y independent and x dependent variable

Q1, Q4, Q5, Q8 similar

Q2, Q3, Q6, Q7 "

General Equation of Ellipse,

$$\frac{x^2}{r_x^2} + \frac{y^2}{r_y^2} = 1$$

Pellipse $\Rightarrow r_y^2 x^2 + r_x^2 y^2 - r_x^2 r_y^2$

(Similar to Circle) Q1 octant

possible ~~pt~~ next pixel (x_k+1, y_k-1)

and (x_k+1, y_k)

Mid-point $\rightarrow \left(\frac{x_k+1+x_k+1}{2}, \frac{y_k-1+y_k}{2}\right) = \left(x_k+1, y_k-\frac{1}{2}\right)$

Now the Pellipse will be,

$$\text{Pellipse} = r_y^2 (x_k+1)^2 + r_x^2 \left(y_k - \frac{1}{2}\right)^2 - r_x^2 r_y^2$$

it is our decision parameter

⇒ As we know, we need initial decision parameter and next decision parameter at each step, So we can get the equations from the main equation after simplifying.

⇒ loop counts:

For Q1 Loop start at $x=0$ and breaks when $m=1$

$$\text{Pellipse} = r_y^2x^2 + r_x^2y^2 - r_x^2r_y^2$$

$$\frac{\delta p}{\delta x} = 2r_y^2x \quad \text{— (I)}$$

$$\frac{\delta p}{\delta y} = 2r_x^2y \quad \text{— (II)}$$

$$\text{(I)} \div \text{(II)} \Rightarrow \frac{\delta y}{\delta x} = \frac{r_y^2x}{r_x^2y}$$

At $m=1$, $r_y^2x = r_x^2y$

So, Loop continues for Q1 $r_y^2x \leq r_x^2y$

and for Q2, $r_x^2y \geq 0$

Algorithm

1. Input r_x, r_y.
2. Initialization $x=0, y=r_y$
3. $P1_0 = r_y^2 - r_x^2 r_y + r_x^2/4$
4. Loop: setpixel $(x,y), (x,-y), (-x,-y), (-x,y)$ with color

   if $(P1_k < 0)$

   $x_{k+1} = x_k + 1$

   $y_{k+1} = y_k$

   $P1_{k+1} = P1_k + 2r_y^2(x_k+1) + r_y^2$

   else $x_{k+1} = x_k + 1$

   $y_{k+1} = y_k - 1$

   $P1_{k+1} = P1_k + 2r_y^2(x_k+1) - 2r_x^2(y_k-1) + r_y^2$

   if $(r_y^2 x_{k+1} \le r_x^2 y_{k+1})$, then go to Loop

   otherwise go to ~~8~~ step 5

5. $P2_0 = r_y^2(x_k + 1/2)^2 + r_x^2(y_k-1)^2 - r_x^2 r_y^2$ where (x_k, y_k) is the last pixel of Q1
6. Loop: setpixel $(x,y), (x,-y), (-x,-y)$ and $(-x,y)$ with color

   if $(P2_k < 0)$

   $x_{k+1} = x_k + 1$

   $y_{k+1} = y_k - 1$

   $P2_{k+1} = P2_k + 2r_y^2(x_k+1) - 2r_x^2 y_k + 3r_x^2$

else

$$x_{k+1} = x_k$$

$$y_{k+1} = y_k - 1$$

$$P2_{k+1} = P2_k - 2r_x^2(y_k - 1) + r_x^2$$

if $(y_{k+1} r_x^2 \geq 0)$ go to Loop

otherwise exit